

International Islamic University Chittagong

Center for General Education (CGED)

Midterm Examination: Spring 2026

Course Code: URIH-4701

Time: 1 hours and 30 minutes.

Program: Undergraduate

Course Title: A Survey of Islamic History & Culture

Full Marks: 30

Instructions:

- All Questions are Compulsory.
- Figures in the right margin indicate full marks.
- Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Bloom's Levels of the Questions.

Letter of Symbol	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

	Text of the Questions	Marks	Bloom's Level	CLO
1	Evaluate the policy of Apostasy (Riddah War) of <i>Khalifah</i> Abu Bakr (R) and point out his significant contributions as the savior of Islam.	10	U	CLO1
2	Review the expansion of Islamic territories during the reign of <i>Khalifah</i> Umar (R). Or Assess the contributions of <i>Khalifah</i> Othman and Ali (R) to the development of Islamic and World Civilization.	10	An	CLO3
3	Write short notes on any two of the followings: a) Election system of four pious <i>Khalifah</i> b) Sources of Revenue under <i>Khalifah</i> Umar (R) c) Development of <i>Shura</i> during four pious <i>Khalifah</i>	10	An	CLO3

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
Mid Term Examination Spring 2026
CSE 4745 Numerical Methods
Total Marks: 30 Time: 90 Minutes

[Answer *all* the *three* questions. Figures in the right-hand margin indicate full marks.]

- 1.a) Write short notes on: 3 CO1
i) Inherent errors ii) Roundoff errors iii) Truncation errors
OR
What do you mean by numerical computing? State four characteristics of numerical computing.
- b) Describe the *banker's rounding rule* with examples. Use *banker's rounding rule* to round off the following numbers to four significant figures - 4 CO1
i) 405.7578 ii) 0.2344500 iii) 0.000098353 iv) 199.9799
- c) A CSE engineer is measuring the length of a wire. The true length of the wire is **125.75162340** meters, but due to instrument limitations, the measured value is **125.68254644** meters. Calculate the *absolute error*, *relative error* and *percentage error*. 3 CO2
- 2.a) What is *Horner's rule*? Evaluate the polynomial $f(x) = 3x^4 - 5x^3 + 4x - 6$ using Horner's rule at $x = M$, where M denotes the *last digit* of your ID number. 2 CO2
- b) Find the root of $f(x) = x^3 - 4x - 9$, correct to *two decimal places*, by using the *False position method*. 5 CO2
OR
Find the root of the equation $f(x) = x^3 - 4x - 9$, correct to two decimal places, by using the *Newton Raphson method*. [Newton-Raphson formula: $X_{n+1} = x_n - f(x_n) / f'(x_n)$]
- c) What is *synthetic division*? Find the *quotient polynomial* $q(x)$ such that $p(x) = (x - 1) q(x)$ where the polynomial $p(x) = x^4 - 6x^3 + 11x^2 - 6x = 0$ has a root at $x = 1$. 3 CO2
- 3.a) What do you mean by *interpolation*? When *Newton's forward* and *Newton's backward* formulas will be used in interpolation and why? 2 CO1
- b) Derive the *Newton's forward interpolation formula*. 4 CO1
OR
Derive the *Newton's divided difference formula*
- c) The following data represents the output of a system: 4 CO1
- | | | | | | |
|----------|----------|-----------|-----------|-----------|-----------|
| x | 2 | 3 | 5 | 7 | 9 |
| y | 6 | 11 | 27 | 51 | 83 |
- Estimate **f(4)** [if the *last digit* of your ID is *odd*] / **f(8)** [if the *last digit* of your ID is *even*] using a *suitable interpolation formula* and justify your choice.

International Islamic University Chittagong
Department of Computer Science and Engineering

B. Sc. in CSE

Mid term Exam, Spring 2026

Course Code: CSE 3633

Course Title: Computer Networks

Time: 1 hour and 30 minutes

Full Marks: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

			C	D	M
1.	a)	Compare and contrast the network models: ISO-OSI, TCP/IP, ATM, and WSN.	1	N	5
1.	b)	Why is the transport layer referred to as the "heart" of the layered architecture in computer networks?	1	U	5
2.	a)	A university campus Wi-Fi system is upgraded to use Slotted ALOHA. Each packet is 1000 bits, data rate is 1 Mbps, and average generation rate is 1200 packets/sec. i. Calculate throughput. ii. Compare with Pure ALOHA. iii. Explain how slotting reduces collision.	2	A	5
2.	b)	Why is the random access technique more suitable than channelization? OR	2	N	5
2	b)	Why is CSMA/CA preferred over CSMA/CD in wireless networks? Explain with flowchart.	2	N	5
3.	a)	Differentiate between datagram and virtual-circuit networks.	1	N	5
3.	b)	What do you mean by subnetting? Given the IP address 192.168.100.0/24, create four subnets of equal size. For each subnet, specify the network address, broadcast address, subnet mask, first usable IP address, and last usable IP address.	2	E	5
		OR			
3	b)	Evaluate CIDR compared to classful routing with examples.	2	E	5

32-24
2

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Spring 2026

Course Code: CSE 4877

Course Title: Machine Learning and Data Mining

Total marks: 30 Time: 1 hour 30 minutes

[Figures in the right-hand margin indicate full marks.]

Ma
rks CO

1.

a) Discuss whether or not each of the following activities is a data mining task. Show justification in one sentence.

2

1

- i. Dividing the customers of a company according to their profitability.
- ii. Monitoring seismic waves for earthquake activities.

b) Suppose you are given a raw dataset for predicting student performance. Explain how the KDD process would be applied to this problem. Discuss each stage from data selection to knowledge discovery.

3

1

OR

b) Why is median often preferred over mean in the presence of outliers?

3

1

c) A dataset contains the attributes Student ID, CGPA, Department, Performance level (Poor, Average, Good, Excellent), Number of absent days. Classify each attribute into its correct type with justification.

5

1

2.

a) Given the dataset:

2, 4, 5, 6, 8, 9, 10, 12, 35

2

2

- i. Find Q_1 , Q_2 (median), and Q_3 .
- ii. Compute the Interquartile Range (IQR).
- iii. Determine the lower and upper bounds for outlier detection using box-plot rule.
- iv. Identify the outlier(s).

b) A class has 50 students, and a researcher wants to select 10 students as a sample.

3

2

- i. What type of numerosity reduction is being used here?
- ii. If every student has an equal chance of being selected, what kind of sampling is this?
Why is this method useful?

c) Consider two document vectors:

1.5

2

- $D_1 = (1, 1)$

+

- $D_2 = (10, 10)$

1.5

+ 2

- i. Compute the cosine similarity between D_1 and D_2 .
- ii. Compute the Euclidean distance between D_1 and D_2 .
- iii. Explain why the two measures lead to different interpretations.

OR

c) Two students are given below: 1.5 2

Student A: Department = CSE, Gender = Female, CGPA = 3.90, Absent Days = 2 +

Student B: Department = CSE, Gender = Female, CGPA = 2.40, Absent Days = 18 1.5 + 2

- i. Are the students similar based on nominal attributes?
- ii. Are they similar based on numeric attributes?
- iii. Explain the overall similarity conceptually for mixed-type data.

a) A dataset contains the attributes- Hours of study, Number of solved problems, Final marks and Student ID. 3 2

- i. Which attributes are likely to be strongly correlated?
- ii. Which attribute is unsuitable for correlation analysis? Why?
- iii. Why is correlation analysis important before model building?

b) A university collected the following student records from different departments to build a machine learning model for predicting the final result. 1+ 2
1+
3

ID	Name	Dept	CGPA	Attendance (%)	Result
101	Maisha	CSE	3.75	85	Pass
102	Ramisa	cse	?	88	Pass
103	Tuli	Computer Science	3.20	92	Pass
104	Nabila	EEE	3.60	30	Pass
105	Jannat	CSE	2.10	10	Fail
106	Sadia	CSE	?	95	Pass
101	Maisha	CSE	3.75	85	Pass
107	Tania	EEE	3.40	99	Pass
108	Fahima	Computer Science	1.90	5	Fail
109	Samia	cse	3.85	100	Pass

- i. Identify the data quality problems present in the above dataset.
- ii. Why data preprocessing is necessary before building a machine learning model for final result prediction?
- iii. Suggest suitable preprocessing steps for each identified problem.

c) Given the data values:

24, 28, 21, 25, 9, 4, 15, 26, 8

Divide them into 3 bins of equal depth.

- i. Smooth the data using bin mean.
- ii. Smooth the data using bin boundaries.

OR

c) Two attributes are given:

- Height (cm): values around 150–180
- Annual Income (BDT): values around 300000–900000

- i. Why may direct comparison of these two attributes be misleading?
- ii. Why is normalization necessary before applying distance-based methods?

International Islamic University Chittagong
Department of Computer Science and Engineering

Mid Term Examination Spring 2026

Course Code: CSE-4741 Course Title: Computer Graphics

Total Marks: 30

Time: 1:30 hours

Answer all questions. Figures in the right margin indicate full marks.

			Mar	CO	DL
1	a)	"The primary goal of computer graphics is to communicate information visually." Do you agree? Explain.	3	CO1	C2
	b)	i. If we use direct coding of RGB values with 3 bits per primary color, how many possible colors do we have for each pixel? ii. If we use 2-bytes pixel values in a 32-bit lookup table representation, how many bytes does the lookup table occupy?	2	CO4	C3
	c)	What do you mean by subtractive color model? Why an additional black pigment is used in printer?	3	CO2	C1
	d)	Compare raster graphics and vector graphics. Which one is more suitable for image synthesis and why?	2	CO2	C1
		OR Write the working principle of CRT monitor? Give necessary figures.			
2	a)	Indicate which raster locations would be chosen by midpoint circle algorithm when scan converting a circle of radius 10 and center at (50, 50). You need to find all raster locations of the circle. OR Indicate which raster location would be chosen by DDA algorithm when scan converting a line from pixel coordinate (3,4) to pixel coordinate (8,10).	5	CO3	C2
	b)	How floating point round-off errors are deducted by decision parameters in Bresenham's line algorithm?	2	CO3	C2
	c)	"Displaying smoothly drawn curves on a pixilated display can produce horribly jagged edges". What are the aliasing effects? How can we solve these problems?	3	CO2	C2
3	a)	i) Find the matrix that represents the rotation of an object by 30° about the origin. ii) What are the new coordinates of the point A(4, -6) after the rotation about B(2, 5)?	4	CO3	C3
	b)	An object is defined in its own coordinate system and needs to be placed into a larger scene. If the object needs to be moved and then rotated, explain why the order of transformation matters .	2	CO1	C2
	c)	Write the general form of a scaling matrix with respect to a fixed point A(h, k).	4	CO3	C3
		OR Find the transformation matrix M_L of mirror reflection of a point about a line L ($y = 2+b$).			

International Islamic University Chittagong
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B. Sc. in CSE Midterm Examination, Spring 2026
Course Code: CSE 4747

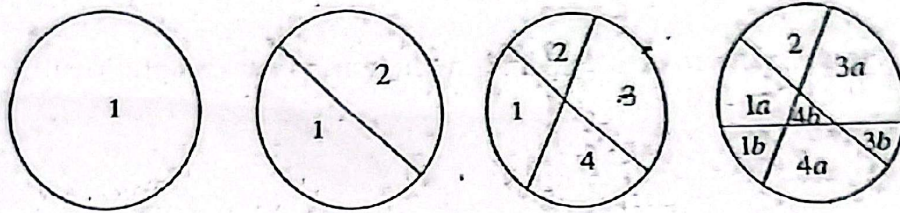
Course Title: Mathematical Analysis for Computer Science

Total marks: 30 Time: 1 hour 30 minutes

[Answer all the questions. Figures in the right-hand margin indicate full marks.]

1. a) As shown in the figure below, suppose that n distinct lines are drawn on a plane such that: 6

- No two lines are parallel.
- No three lines intersect at the same point.



Let $R(n)$ denote the minimum number of regions into which the plane is divided by these n lines.

- i) Write the recurrence relation for $R(n)$.
 - ii) Solve the recurrence to obtain a closed-form expression for $R(n)$.
 - iii) Find the value of $R(9)$.
- b) Use the Akra-Bazzi formula to find the asymptotic bounds for the following recurrence: 4

$$T(n) = 4T(n/2) + n^2, \quad T(1) = 1 \text{ and } T(n) = \Theta(1)$$

2. a) Evaluate the following expression: 2

$$\sum_{k=17}^{25} k^3$$

- b) Find a closed form for the following expression: 3

i) $\sum_{k=1}^{10} \sum_{j=1}^k \sum_{i=1}^j 1$ ii) $\sum_{i=0}^n \sum_{j=0}^i \sum_{k=0}^j \left(\frac{p}{q}\right)^k$ + 3

- c) Compute $J(24)$ using both the recurrent and binary properties of the Josephus problem while eliminating every 2nd person. 2

3.

a) Assume that you are organizing a university tech conference and need to form a committee of n members from different departments. Use generating functions to determine in how many ways a committee of n members can be formed under the following constraints:

5

- o The number of faculty advisors is at most 1
- o The number of technical coordinators is at most 2
- o The number of marketing members must be even
- o The number of event manager is at most 1
- o The number of volunteers must be a multiple of 3
- o The number of design team members must be even

Using generating function, find the number of possible committees when $n = 8$.

b) Write down a faster alternative to the following code segment, i.e. optimize it:

2

```
int N = 100;  
float sum = 0;  
for (int i = 1; i <= N; i++)  
    sum += i * (i + 1);
```

c) If a recurrence relation keeps calling itself to solve its problems, is it doing dynamic programming or just avoiding responsibility? Explain your answer with reasoning. 1.5

d) If a generating function encodes an entire sequence inside itself, is it the most organized student in class or just hoarding information? Explain your answer with reasoning. 1.5